

Unit 2: Big Data Infrastructures (Compute/Storage) — **Question Bank**

2 Mark Questions

Topics covered: Hadoop Ecosystem, Apache Hive, Apache HBase, Apache Cassandra, Apache Iceberg, Apache SeaTunnel, Apache Druid, Apache Hudi, Apache Airflow

50 Questions

(Q51–Q80)

Q51. *[MCQ; 2 marks]*

What is the correct sequence of phases in a classic MapReduce job?

- A)** Map, then Shuffle/Sort, then Reduce.
- B)** Reduce, then Map, then Shuffle.
- C)** Reduce, then Shuffle, then Map.
- D)** Shuffle, then Map, then Reduce.

Ans:

The correct answer is A — MapReduce first maps input into intermediate key-value pairs, then shuffles/sorts them by key, then reduces grouped values into the final output.

Q52. *[Fill in the blank; 2 marks]*

The Hadoop daemon that runs on every worker node, manages local containers, and reports resource usage to the ResourceManager is called the _____.

Ans:

The correct answer is "NodeManager" — each worker node runs a NodeManager, which manages containers locally and reports available CPU and memory back to the cluster-wide ResourceManager.

Q53. *[MCQ; 2 marks]*

How does partitioning in Apache Hive primarily improve query performance?

- A)** It enforces foreign-key relationships between tables.
- B)** It encrypts sensitive table data before it is written.
- C)** It lets queries skip scanning irrelevant partitions.
- D)** It automatically generates charts from query results.

Ans:

The correct answer is C — partitioning divides a Hive table's data (for example, by date) into separate directories, so queries filtering on the partition column can skip unrelated data.

Q54. *[Fill in the blank; 2 marks]*

In Hive, a table whose data lives outside the default warehouse directory and is NOT deleted when the table is dropped is called an _____ table.

Ans:

The correct answer is "external" — external tables only let Hive manage the metadata, while the actual data continues to exist at its original location even after the table is dropped.

Q55. *[MCQ; 2 marks]*

How is a single "cell" of data uniquely identified within an HBase table?

- A) By a foreign key referencing a related table.
- B) By a single global auto-incrementing primary key alone.
- C) By row key, column family:qualifier, and timestamp.
- D) By table name and an index number only.

Ans:

The correct answer is C — an HBase cell is uniquely addressed by its row key, its column family:qualifier, and the timestamp/version associated with that write.

Q56. *[Fill in the blank; 2 marks]*

HBase tables are horizontally split into contiguous row-key ranges called _____, each of which is served by exactly one RegionServer at a time.

Ans:

The correct answer is "regions" — regions are the unit of horizontal scalability in HBase, distributed across RegionServers to allow the table to scale to very large sizes.

Q57. [MCQ; 2 marks]

Which of the following best reflects a core design goal of Apache Cassandra?

- A)** Relying on a single powerful master node to simplify coordination.
- B)** Supporting only vertical scaling via hardware upgrades.
- C)** Enforcing strict two-phase commit for every single write.
- D)** High availability with no single point of failure.

Ans:

The correct answer is D — Cassandra was explicitly designed for high availability and linear horizontal scalability, avoiding any single point of failure through its masterless architecture.

Q58. [Fill in the blank; 2 marks]

In Cassandra, the set of columns used to determine which node(s) will store a given row of data is called the _____ key.

Ans:

The correct answer is "partition" — the partition key is hashed to determine the token, and therefore the node(s), responsible for storing that row on the Cassandra ring.

Q59. [MCQ; 2 marks]

Which types of schema changes does Apache Iceberg support safely, without requiring existing data files to be rewritten?

- A)** Only changing the underlying file format used.
- B)** Only adding brand-new columns, with no other changes.
- C)** Only deleting the entire table and recreating it.
- D)** Adding, dropping, renaming, and reordering columns.

Ans:

The correct answer is D — Iceberg's schema evolution allows a wide range of safe changes, including adding, dropping, renaming, and reordering columns, without rewriting data files.

Q60. *[Fill in the blank; 2 marks]*

Apache _____ is described in the unit syllabus as a very high-performance distributed data integration tool used for synchronizing data between hundreds of data sources.

Ans:

The correct answer is "SeaTunnel" — this directly matches SeaTunnel's stated design purpose of large-scale, distributed data synchronization across many heterogeneous systems.

Q61. *[MCQ; 2 marks]*

Which workload is Apache Druid most suitable for handling?

- A)** Traditional OLTP banking transactions with row-level locking.
- B)** Exclusively nightly batch reporting with no live data support.
- C)** Interactive, low-latency OLAP analytics over event data.
- D)** Simple static file storage with no query capability.

Ans:

The correct answer is C — Druid is purpose-built for fast, interactive analytical queries over time-series and event-style data, whether real-time or historical.

Q62. *[Fill in the blank; 2 marks]*

In Apache Druid, the immutable, columnar chunks of indexed data that form the core unit of storage are called _____.

Ans:

The correct answer is "segments" — Druid organizes ingested data into immutable segments, optimized for compression and fast analytical scanning.

Q63. [MCQ; 2 marks]

Which operation does Apache Hudi make efficient that is otherwise difficult to perform on plain, immutable Parquet files in a data lake?

- A)** Storing data using a columnar file format.
- B)** Efficient record-level upserts and deletes without a full rewrite.
- C)** Simply reading previously written data from the files.
- D)** Compressing files to reduce their on-disk footprint.

Ans:

The correct answer is B — plain immutable files typically require rewriting large portions of data for any change, whereas Hudi enables targeted, efficient upserts and deletes at the record level.

Q64. [Fill in the blank; 2 marks]

Apache _____ is the tool in this unit's syllabus that specifically brings real-time updates and incremental processing capabilities to data lakes.

Ans:

The correct answer is "Hudi" — this is Hudi's defining capability, enabling upserts, deletes, and incremental data consumption on top of data lake storage.

Q65. [MCQ; 2 marks]

Which Apache Airflow component continuously parses DAG files and decides when tasks are ready to be triggered?

- A)** The Metadata Database, which only passively stores state.
- B)** The Worker process, which only executes triggered tasks.
- C)** The Scheduler, which parses DAGs and triggers tasks.
- D)** The Webserver, which only serves the user interface.

Ans:

The correct answer is C — the Scheduler is the component that continuously monitors DAG definitions and schedules, deciding when each task instance should be triggered.

Q66. *[Fill in the blank; 2 marks]*

In Apache Airflow, a single execution of an entire DAG at a specific logical date/time is referred to as a _____.

Ans:

The correct answer is "DAG Run" — each scheduled or manually triggered execution of the full workflow is tracked in Airflow as an individual DAG Run.

Q67. *[MCQ; 2 marks]*

Which statement best captures the relationship between Apache Hive and the underlying Hadoop platform?

- A)** The core Hadoop framework actually runs as a component inside Hive.
- B)** Hive and Hadoop are entirely unrelated systems.
- C)** Hive completely replaces Hadoop, requiring no other components.
- D)** Hive is a SQL-like abstraction layer built on Hadoop's storage and compute.

Ans:

The correct answer is D — Hive provides a familiar SQL-like interface (HiveQL) over data that is physically stored and processed using Hadoop's underlying storage and compute engines.

Q68. *[Fill in the blank; 2 marks]*

The default block size commonly used in HDFS is _____ MB.

Ans:

The correct answer is "128" — HDFS commonly uses a default block size of 128 MB, although older versions historically defaulted to 64 MB.

Q69. *[MCQ; 2 marks]*

Which statement about column families in HBase is correct?

- A)** They must be declared dynamically for every single write.
- B)** They entirely eliminate the need for a row key.
- C)** They map one-to-one with individual rows in the table.
- D)** They are declared upfront at table creation and remain relatively static.

Ans:

The correct answer is D — column families are defined when a table is created and remain relatively static, although individual column qualifiers can be added dynamically at write time.

Q70. *[Fill in the blank; 2 marks]*

Cassandra's query language, which is syntactically similar to standard SQL, is called _____.

Ans:

The correct answer is "CQL" (Cassandra Query Language) — CQL provides a familiar SQL-like syntax for defining and querying tables in Cassandra.

Q71. *[MCQ; 2 marks]*

What is the main purpose of a "catalog" in the context of Apache Iceberg tables?

- A)** To physically store all of the actual table data rows.
- B)** To map table names to their current metadata file location.
- C)** To manage the replication of HDFS blocks across DataNodes.
- D)** To schedule when Airflow DAGs should be triggered.

Ans:

The correct answer is B — an Iceberg catalog maps each table name to the location of its current metadata file, and atomically updating this pointer is what makes commits safe and consistent.

Q72. *[MCQ; 2 marks]*

What best characterizes the connector-based design that lets SeaTunnel synchronize "huge amounts of data" across many systems?

- A)** A requirement that every source and sink share one database engine.
- B)** A manual, entirely GUI-driven process with no configuration files.
- C)** A single hard-coded script written separately for every source-sink pair.
- D)** A pluggable Source-Transform-Sink framework reused across systems.

Ans:

The correct answer is D — SeaTunnel's pluggable connector framework, structured around Source-Transform-Sink, is what allows it to scale to large-volume synchronization across many heterogeneous systems.

Q73. [MCQ; 2 marks]

Which of the following is NOT a typical node/process role within a Druid cluster?

- A)** NameNode, which is actually an HDFS component, not a Druid role.
- B)** Coordinator, which manages segment distribution and balancing.
- C)** Historical, which stores and serves queries on finalized segments.
- D)** Broker, which routes and merges query results.

Ans:

The correct answer is A — the NameNode belongs to HDFS's architecture, not Druid's; Druid's own roles instead include Coordinator, Overlord, Broker, Historical, and MiddleManager.

Q74. [Fill in the blank; 2 marks]

Hudi's timeline mechanism records every action performed on a table, such as commits, compactions, and cleans, which enables consumers to perform _____ processing of only the changed records.

Ans:

The correct answer is "incremental" — by tracking every commit on its timeline, Hudi lets downstream consumers pull only the records that changed since a given point, rather than reprocessing the entire dataset.

Q75. [MCQ; 2 marks]

Which Airflow feature allows a task to wait and repeatedly check for an external condition, such as a file's arrival, before letting downstream tasks proceed?

- A)** A standard Operator, which always runs immediately.
- B)** A Sensor, a special operator designed to poll for a condition.
- C)** An Executor, which only decides where tasks run.
- D)** A Hook, which is only used to connect to external systems.

Ans:

The correct answer is B — Sensors are specialized operators that repeatedly poll for a condition (for example, a file appearing in storage) to become true before allowing downstream tasks to execute.

Q76. *[Fill in the blank; 2 marks]*

Within the Hadoop ecosystem, the component that arbitrates and allocates cluster resources among multiple competing applications is called _____.

Ans:

The correct answer is "YARN" — YARN acts as the resource management layer, allocating CPU and memory across applications like MapReduce, Spark, and Hive that share the same cluster.

Q77. *[MCQ; 2 marks]*

Which statement about views in Apache Hive is correct?

- A)** Views physically duplicate and permanently store the underlying data.
- B)** Views are not permitted to reference any other tables.
- C)** Views can only be created if the storage engine is HBase.
- D)** Views are virtual tables defined by a query, storing no data.

Ans:

The correct answer is D — a Hive view is a logical, virtual table backed by a stored query definition; it does not physically store its own copy of the data.

Q78. *[Fill in the blank; 2 marks]*

In HBase, the in-memory structure where recent writes to a column family are buffered before being flushed to disk as an HFile is called the _____.

Ans:

The correct answer is "MemStore" — the MemStore holds recently written data in memory per column family before it is flushed to a persistent, immutable HFile on disk.

Q79. *[MCQ; 2 marks]*

What does Cassandra's "NetworkTopologyStrategy" replication strategy primarily allow administrators to do?

- A)** Restrict the cluster to running on a single node only.
- B)** Disable replication entirely across the cluster.
- C)** Completely ignore data center topology when placing replicas.
- D)** Specify a different replication factor per data center.

Ans:

The correct answer is D — NetworkTopologyStrategy lets administrators configure a separate replication factor per data center, especially useful for multi-data-center deployments.

Q80. *[Fill in the blank; 2 marks]*

Apache _____ is described in the syllabus as a high-performance table format for data lakes that provides SQL-like ACID transactions and schema evolution.

Ans:

The correct answer is "Iceberg" — this description directly matches the definition of Apache Iceberg as introduced in the unit syllabus.

(Q81–Q93)

Q81. *[MCQ; 2 marks]*

What is the primary purpose of "rack awareness" in HDFS block placement?

- A) To assign static IP addresses to DataNodes.
- B) To place replicas across racks, balancing fault tolerance and bandwidth.
- C) To automatically encrypt every block before writing to disk.
- D) To decide which Airflow DAGs run on which cluster nodes.

Ans:

The correct answer is B — rack-aware placement spreads replicas of a block across different racks, so a single rack failure does not lose all copies, while limiting cross-rack network traffic.

Q82. *[Fill in the blank; 2 marks]*

In Hive, converting row-oriented data into a columnar storage format such as ORC or _____ can substantially improve the performance of analytical queries.

Ans:

The correct answer is "Parquet" — both ORC and Parquet are columnar storage formats commonly paired with Hive to speed up analytical scans by reading only the required columns.

Q83. [MCQ; 2 marks]

Which consistency guarantee does HBase provide by default for a single-row operation?

- A)** Strong, atomic consistency for reads and writes on a single row.
- B)** No consistency guarantee is provided at all.
- C)** Consistency only if a two-phase commit is manually triggered.
- D)** Only eventual consistency, similar to Cassandra's default.

Ans:

The correct answer is A — HBase guarantees strong, atomic consistency for operations performed on a single row, even though it does not guarantee this across multiple rows in one operation.

Q84. [Fill in the blank; Numerical; 2 marks]

In a Cassandra cluster with a replication factor of 3, if a write is performed using the QUORUM consistency level, at least _____ replicas must acknowledge the write before it is considered successful.

Ans:

The correct answer is "2" — QUORUM requires acknowledgment from a majority of replicas, calculated as $\text{floor}(\text{RF}/2) + 1$; with $\text{RF} = 3$, that works out to $\text{floor}(3/2) + 1 = 2$.

Q85. [MCQ; 2 marks]

What is primarily stored inside an Apache Iceberg manifest file?

- A)** User authentication credentials required to access the table.
- B)** Scheduling information describing when Airflow DAGs run.
- C)** A list of data files with partition values and column statistics.
- D)** The raw rows of table data themselves.

Ans:

The correct answer is C — manifest files list the data files that make up a snapshot, along with statistics such as partition values and column ranges, used to prune irrelevant files.

Q86. *[Fill in the blank; 2 marks]*

SeaTunnel's data pipelines are typically structured around three stages: Source, _____, and Sink.

Ans:

The correct answer is "Transform" — like most data integration tools, SeaTunnel pipelines follow a Source (read) → Transform (process) → Sink (write) model.

Q87. *[MCQ; 2 marks]*

What is the main purpose of performing "roll-up" during data ingestion in Apache Druid?

- A)** To pre-aggregate rows sharing the same dimensions, reducing storage.
- B)** To encrypt streaming data before it is written to storage.
- C)** To convert incoming JSON records into XML format.
- D)** To automatically delete old segments once they reach a certain age.

Ans:

The correct answer is A — roll-up aggregates rows that share identical dimension values and timestamp granularity at ingestion time, reducing stored rows and speeding up queries.

Q88. *[Fill in the blank; 2 marks]*

In Apache Hudi, the _____ table type stores updates as delta log files that are only merged with the base files at read time or during compaction.

Ans:

The correct answer is "Merge-on-Read (MOR)" — MOR tables defer the cost of merging updates until read time or a background compaction process, unlike Copy-on-Write tables.

Q89. [MCQ; 2 marks]

What does "backfilling" refer to in the context of Apache Airflow?

- A)** Permanently deleting old DAG run records from the database.
- B)** Running a DAG for past scheduled intervals that were missed.
- C)** Restarting the Airflow webserver after a configuration change.
- D)** Creating a backup copy of the Airflow metadata database.

Ans:

The correct answer is B — backfilling executes DAG runs corresponding to historical schedule intervals, useful for reprocessing data or catching up on missed runs.

Q90. [Fill in the blank; 2 marks]

The part of Hive responsible for parsing a HiveQL statement and producing a logical and physical execution plan is generally referred to as the query _____, within Hive's Driver component.

Ans:

The correct answer is "compiler" — Hive's Driver includes a compiler and optimizer that parse HiveQL statements and generate an execution plan suited to the chosen execution engine.

Q91. [MCQ; 2 marks]

Which scenario provides the strongest justification for choosing HBase over Hive?

- A)** The team primarily needs a tool to orchestrate scheduled workflows.
- B)** The application requires real-time OLAP-style dashboard analytics.
- C)** The application needs low-latency random reads/writes at scale.
- D)** The team needs ad-hoc SQL analytics over historical batch data.

Ans:

The correct answer is C — HBase is specifically optimized for fast, random access to individual records, a very different access pattern from Hive's batch-oriented analytical queries.

Q92. *[Fill in the blank; 2 marks]*

Cassandra uses a probabilistic data structure called a _____ filter to quickly check whether an SSTable might contain a given row key before performing an actual disk read.

Ans:

The correct answer is "Bloom" — Bloom filters allow Cassandra to skip SSTables that definitely do not contain a requested key, reducing unnecessary disk I/O.

Q93. *[MCQ; 2 marks]*

What primarily makes "time travel" queries possible in Apache Iceberg?

- A)** The use of real-time streaming ingestion alone.
- B)** HDFS block replication, which is unrelated to Iceberg's design.
- C)** Airflow's scheduling capabilities, controlling query execution.
- D)** Retained historical snapshot metadata recording table state over time.

Ans:

The correct answer is D — because Iceberg retains snapshots and their associated metadata over time, a query can be directed at a previous snapshot or timestamp.

(Q94–Q100)

Q94. *[MCQ; 2 marks]*

How does the NameNode typically detect that a DataNode in the cluster has failed?

- A)** The NameNode has no mechanism to detect DataNode failures.
- B)** Through notifications sent exclusively by the YARN ResourceManager.
- C)** Only through a manual command issued by an administrator.
- D)** Through periodic heartbeats; absence within a timeout marks it dead.

Ans:

The correct answer is D — DataNodes send periodic heartbeats to the NameNode, and if a heartbeat is not received within a configured timeout, the NameNode marks that DataNode as dead and triggers re-replication.

Q95. *[Fill in the blank; 2 marks]*

When a RegionServer crashes in HBase, the HMaster reassigns its regions to other RegionServers by replaying the _____, which contains unflushed edits for durability.

Ans:

The correct answer is "WAL" (Write-Ahead Log) — the WAL ensures durability by recording writes before they are applied to the MemStore, allowing unflushed edits to be recovered by replaying the log after a crash.

Q96. *[MCQ; 2 marks]*

What is the purpose of Cassandra's "hinted handoff" mechanism?

- A)** To permanently remove unreachable nodes from the cluster's ring.
- B)** To balance CPU load evenly across all nodes in the cluster.
- C)** To temporarily store writes meant for a down node, replayed on recovery.
- D)** To encrypt data as it travels from the client to the coordinator.

Ans:

The correct answer is C — hinted handoff lets a coordinator node temporarily hold a "hint" representing a write meant for a down replica, replaying that write once the replica becomes available again.

Q97. *[Fill in the blank; 2 marks]*

Apache Iceberg safely supports concurrent writers by relying on _____ concurrency control at the level of the catalog's metadata pointer.

Ans:

The correct answer is "optimistic" — Iceberg uses optimistic concurrency control, where a writer commits by atomically swapping the catalog's metadata pointer only if it has not changed since it was last read, retrying otherwise.

Q98. [MCQ; 2 marks]

Why are segments in Apache Druid typically designed to be immutable once they are created?

- A)** Immutability simplifies caching, replication, and parallel scanning.
- B)** Immutability is used purely to reduce network bandwidth.
- C)** Druid is fundamentally incapable of supporting any data update.
- D)** The underlying HDFS filesystem strictly enforces immutability.

Ans:

The correct answer is A — treating segments as immutable simplifies distributed caching and replication, and allows Druid to scan segments in parallel without concurrent-modification concerns.

Q99. [Fill in the blank; 2 marks]

In Apache Hudi, the record-level _____ maps a record's key to the file group in which it resides, allowing efficient upserts without scanning the entire table.

Ans:

The correct answer is "index" — Hudi's indexing mechanism (for example, a Bloom index) maps record keys to the file groups containing them, so upserts can locate existing records quickly.

Q100. [MCQ; 2 marks]

What is the main risk of placing heavy, long-running computations as top-level code directly inside an Airflow DAG file, outside of any operator?

- A)** It becomes a mandatory requirement for XComs to function.
- B)** Top-level code in a DAG file has no meaningful effect at all.
- C)** It automatically transforms that code into a Sensor.
- D)** It re-executes on every Scheduler parse cycle, slowing scheduling.

Ans:

The correct answer is D — because the Scheduler repeatedly parses DAG files to detect changes, heavy top-level computation executes on every parse cycle, which can seriously degrade scheduler performance.

Answer Key

Section B (Q51-Q100): 51-A, 52-NodeManager, 53-C, 54-external, 55-C, 56-regions, 57-D, 58-partition, 59-D, 60-SeaTunnel, 61-C, 62-segments, 63-B, 64-Hudi, 65-C, 66-DAG Run, 67-D, 68-128, 69-D, 70-CQL, 71-B, 72-D, 73-A, 74-incremental, 75-B, 76-YARN, 77-D, 78-MemStore, 79-D, 80-Iceberg, 81-B, 82-Parquet, 83-A, 84-2, 85-C, 86-Transform, 87-A, 88-Merge-on-Read (MOR), 89-B, 90-compiler, 91-C, 92-Bloom, 93-D, 94-D, 95-WAL, 96-C, 97-optimistic, 98-A, 99-index, 100-D